

CS766 Computer Vision Final Project

Project Proposal

Team Members:

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The problem:

There is race footage (<https://www.youtube.com/@CMUbuggy>) that we would like to analyze. It has pans and zooms within shots and cuts between different camera angles. We intend to extract both 3-D (course environment) and 4-D (racer progress) data.

The same techniques that will work on this example are broadly applicable to geolocation and object tracking more generally. There is significant amounts of labeled video footage available in this case.

State of the art:

There are existing libraries to accomplish similar tasks, but we do not have experience with them. We intend to gain experience through this project. There is no existing general solution to automatically extract location data and object positioning from video footage.

We will use existing work where possible, and develop new approaches where necessary. We do not yet know where new approaches will be necessary.

End product:

The initial result is a topographical map of the course environment (the road, trees, and other features). The secondary goal is adding tracks by race and year. The stretch goal end result for presentation purposes would be a composited video showing several races merged into one.

Timeline:

March 6: Video acquisition and export for use in a code friendly form.

March 20: Initial environment feature extraction complete.

April 10: Racer data extraction complete. Final refinements started.

April 20: Final Presentation complete.